



Grade 8 SS Geography

3. Climate Around the World

Learning Guide

Introduction

- **Climate varies around the world** because different places have different conditions.
- Climate is affected by:
 - **Location**
 - **Altitude**
 - **Distance from the sea**
 - **Ocean currents**
- Scientists group climates into major types based on **temperature** and **rainfall patterns**.

What is a Climate Region?

- A **climate region** is an area of the world where **similar weather patterns** occur over a long period.
- Different climate regions have different:
 - **Temperatures**
 - **Rainfall**
 - **Weather patterns**
- Climate regions are influenced by:
 - **Latitude**
 - **Altitude**
 - **Distance from the sea**
 - **Ocean currents**
 - **Mountains / Relief**

Major Climate Regions of the World

1. Tropical Climate

- Found near the **Equator**.
- It is **hot all year round**.
- It receives **heavy rainfall**, especially in summer.
- **Dense forests**, such as **rainforests**, are common in this climate region.

2. Subtropical Climate

- Found just beyond the **tropical zones**.
- Has **hot summers** and **mild winters**.
- Rain often comes from **summer thunderstorms**.

3. Temperate Climate

- Found in **mid-latitude areas**.
- Has **mild summers** and **cool winters**.
- Receives **regular rainfall throughout the year**.

4. Desert Climate

- **Hot and dry** with very little rainfall.

- Has a large difference between **day and night temperatures**.
- Vegetation is **sparse**, meaning there are few plants.

5. Semi-Desert Climate

- Similar to a **desert climate**, but receives **slightly more rainfall**.
- Supports some **shrubs and grasses**.

6. Continental Climate

- Found in large continents **away from the ocean**.
- Has **hot summers** and **very cold winters**.
- Has a **wide temperature range** because the ocean does not moderate the temperature.

7. Polar Climate

- Found near the **poles**, such as the **Arctic** and **Antarctic**.
- It is **extremely cold all year**.
- It receives **very little precipitation**, mostly in the form of **snow**.

8. Mediterranean Climate

- Has **hot, dry summers** and **mild, wet winters**.
- Found around the **Mediterranean Sea** and in parts of the **Western Cape** in South Africa.

9. Tundra Climate

- Found near the **Arctic Circle**.
- It is **very cold most of the year**.
- Has **short, cool summers**.
- **Permafrost**, meaning frozen ground, is common.

10. High Mountain Climate

- Found in **mountain ranges**.
- Similar to a **polar climate** because of high altitude.
- It is **cold**, and snow may remain on mountain peaks even in summer.

Bar and Line Graphs

- **Bar and line graphs** are used to show **climate data**.
- A **bar graph** shows **rainfall**.
- A **line graph** shows **temperature**.

How to Read a Climate Region Map

When studying a climate region map, notice the following:

- **Tropical regions** are near the **Equator**.
 - **Polar regions** are near the **poles**.
 - **Desert regions** are often found around **30° north and south** of the Equator.
 - **Temperate regions** are usually between **30° and 60° north and south**.
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Link Between Climate Regions and Factors Affecting Temperature and Rainfall

1. Latitude

- Regions closer to the **Equator**, such as **tropical regions**, receive **direct sunlight** and are therefore **hot**.
- Regions near the **poles**, such as **polar regions**, receive **slanted rays** and are therefore **cold**.

2. Distance from the Sea

- **Coastal regions**, such as Mediterranean areas, have **moderate temperatures**.
- **Inland regions**, such as continental areas, have **extreme temperatures**.

3. Altitude

- **High mountain regions** are cold because of their **high altitude**.
- Example: **Himalayas**.

4. Ocean Currents

- **Warm ocean currents**, such as the **Agulhas Current**, create **warm coastal climates**.
- **Cold ocean currents**, such as the **Benguela Current**, create **cooler coastal climates**.

5. Relief / Mountains

- Mountains can block rain and create **wet and dry sides**.
- Example: The **Andes Mountains** affect the climates of South America.

Examples of Climate Regions and Influencing Factors

Region	Climate Type	Influencing Factors
Amazon Basin	Tropical	Close to the Equator, low altitude, inland but with high rainfall
Sahara Desert	Desert	Latitude around 30°N, subtropical high-pressure belt, distance from the sea
Cape Town	Mediterranean	Proximity to the sea, latitude, ocean currents
Himalayas / Drakensberg	High Mountain	Very high altitude

✓ Key Takeaways

- ✓ Climate differs around the world because of **location, altitude, distance from the sea, ocean currents, and relief**.
- ✓ **Tropical climates** are hot and wet because they are near the Equator.
- ✓ **Desert climates** are hot and dry with little rainfall.
- ✓ **Polar and tundra climates** are very cold.
- ✓ **Mediterranean climates** have hot, dry summers and mild, wet winters.
- ✓ **High mountain climates** are cold because of altitude.
- ✓ **Bar graphs** show rainfall, while **line graphs** show temperature.

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